

Problem 1

Let E_{ij} be the event that person i and person j become close friends, E_{ij}^c be the event that they argue, and V_{ij} be the event that information (the gossip) from i reaches j . We are given that $\mathbb{P}(E_{ij}^c) = p$ and $\mathbb{P}(E_{ij}) = 1 - p =: q, \forall i, j \in \{A, B, C, D\}, i \neq j$ where A, B, C, D denote Anne, Betty, Chloë, and Daisy. Figure 1 illustrates the graph model of the problem.

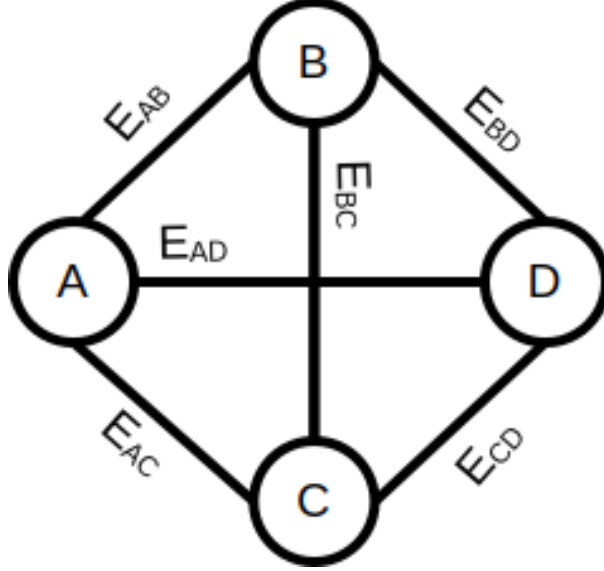


Figure 1: Graph model of the problem. The vertices A, B, C , and D represent Anne, Betty, Chloë, and Daisy, respectively. The edges represent the potential friendships between them. Each edge exists with probability q and does not exist (argument) with probability p .

Additionally, let $\Omega = \{0, 1\}^6$, where $\omega = (\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6) \in \Omega$, and $\omega_k = 1$ if the k -th edge in $H := (AB, AC, AD, BC, BD, CD)$ exists, and 0 otherwise. Because the arguments occur independently, the probability measure $\mathbb{P}$ on $(\Omega, \mathcal{P}(\Omega))$ is the product measure, determined by $\mathbb{P}(E_{H_k}) = \mathbb{P}(\{\omega \in \Omega \mid \omega_k = 1\}) = q$ and $\mathbb{P}(E_{H_k}^c) = \mathbb{P}(\{\omega \in \Omega \mid \omega_k = 0\}) = p$.

Part a)

We want to find $\mathbb{P}(V_{AD} \mid E_{AD}^c)$. Without the direct connection AD , the remaining graph consists of edges $\{AB, AC, BD, CD, BC\}$. We note that the presence or absence of any edge forms a partition of the sample space, i.e., $\Omega = E_{H_k} \dot{\cup} E_{H_k}^c = \{\omega \in \Omega \mid \omega_k = 1\} \dot{\cup} \{\omega \in \Omega \mid \omega_k = 0\}$. We apply the Law of Total Probability on the partition created by E_{BC} and E_{BC}^c :

$$\mathbb{P}(V_{AD} \mid E_{AD}^c) = \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC})\mathbb{P}(E_{BC} \mid E_{AD}^c) + \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}^c)\mathbb{P}(E_{BC}^c \mid E_{AD}^c)$$

Since E_{AD} is independent of E_{BC} , and independence carries over their complements (Lemma 2.2.4), we have $\mathbb{P}(E_{BC} \mid E_{AD}^c) = \mathbb{P}(E_{BC}) = q$ and $\mathbb{P}(E_{BC}^c \mid E_{AD}^c) = \mathbb{P}(E_{BC}^c) = p$. Thus, we can simplify the above expression to:

$$\begin{aligned} \mathbb{P}(V_{AD} \mid E_{AD}^c) &= \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}) \cdot \mathbb{P}(E_{BC}) + \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}^c) \cdot \mathbb{P}(E_{BC}^c) \\ &= \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}) \cdot q + \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}^c) \cdot p \end{aligned} \quad (1)$$

We analyze the two terms separately:

- Given that E_{BC} occurs (left term with probability q), the event V_{AD} occurs if A connects to $\{B, C\}$ and $\{B, C\}$ connects to D . Because the remaining edges are independent of the conditioned edges E_{AD} and E_{BC} we can drop the conditioning:

$$\mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}) = \mathbb{P}((E_{AB} \cup E_{AC}) \cap (E_{BD} \cup E_{CD}))$$

Because the edge sets $\{AB, AC\}$ and $\{BD, CD\}$ are disjoint, the σ -algebras generated by them are independent (Lemma 2.2.11). Thus, we can multiply their probabilities:

$$\begin{aligned}\mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}) &= \mathbb{P}(E_{AB} \cup E_{AC}) \cdot \mathbb{P}(E_{BD} \cup E_{CD}) \\ &= (1 - \mathbb{P}(E_{AB}^c \cap E_{AC}^c)) \cdot (1 - \mathbb{P}(E_{BD}^c \cap E_{CD}^c)) \\ &= (1 - \mathbb{P}(E_{AB}^c)\mathbb{P}(E_{AC}^c)) \cdot (1 - \mathbb{P}(E_{BD}^c)\mathbb{P}(E_{CD}^c)) \\ &= (1 - p \cdot p)(1 - p \cdot p) = (1 - p^2)^2\end{aligned}$$

- Given that E_{BC}^c occurs (right term with probability p), the graph reduces to two parallel, independent paths, $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$. Again, by Lemma 2.2.11, the paths rely on disjoint sets of edges and are therefore independent:

$$\begin{aligned}\mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}^c) &= \mathbb{P}((E_{AB} \cap E_{BD}) \cup (E_{AC} \cap E_{CD})) \\ &= 1 - \mathbb{P}((E_{AB} \cap E_{BD})^c) \cdot \mathbb{P}((E_{AC} \cap E_{CD})^c) \\ &= 1 - (1 - q^2)^2\end{aligned}\tag{2}$$

Substituting the above results into equation 1, we obtain:

$$\mathbb{P}(V_{AD} \mid E_{AD}^c) = q(1 - p^2)^2 + p(1 - (1 - q^2)^2)\tag{3}$$

Part b)

We apply the Law of Total Probability by partitioning the sample space on the direct edge E_{AD} and its complement E_{AD}^c :

$$\mathbb{P}(V_{AD}) = \mathbb{P}(V_{AD} \mid E_{AD})\mathbb{P}(E_{AD}) + \mathbb{P}(V_{AD} \mid E_{AD}^c)\mathbb{P}(E_{AD}^c)$$

The event V_{AD} is the union of all paths from A to D . Therefore, $E_{AD} \subset V_{AD}$, and $\mathbb{P}(V_{AD} \mid E_{AD}) = \frac{\mathbb{P}(V_{AD} \cap E_{AD})}{\mathbb{P}(E_{AD})} = \frac{\mathbb{P}(E_{AD})}{\mathbb{P}(E_{AD})} = 1$. Substituting this, and equation 3, we obtain:

$$\begin{aligned}\mathbb{P}(V_{AD}) &= 1 \cdot q + \mathbb{P}(V_{AD} \mid E_{AD}^c) \cdot p \\ &= q + p [q(1 - p^2)^2 + p(1 - (1 - q^2)^2)]\end{aligned}$$

Part c)

We apply the Law of Total Probability by partitioning the space on the direct edge E_{AD} and its complement E_{AD}^c :

$$\mathbb{P}(V_{AD} \mid E_{BC}^c) = \mathbb{P}(V_{AD} \mid E_{AD} \cap E_{BC}^c)\mathbb{P}(E_{AD} \mid E_{BC}^c) + \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{BC}^c)\mathbb{P}(E_{AD}^c \mid E_{BC}^c)\tag{4}$$

By the independence of the individual edges, we have $\mathbb{P}(E_{AD} \mid E_{BC}^c) = \mathbb{P}(E_{AD}) = q$ and $\mathbb{P}(E_{AD}^c \mid E_{BC}^c) = \mathbb{P}(E_{AD}^c) = p$. Furthermore, as established in part b), $\mathbb{P}(V_{AD} \mid E_{AD}) = 1$, and, therefore, $\mathbb{P}(V_{AD} \mid E_{AD} \cap E_{BC}^c) = 1$. If E_{AD}^c occurs, the network lacks both the AD and BC edges, which is exactly the parallel-path scenario evaluated in equation 2. Substituting all of these values into equation 4, we obtain:

$$\mathbb{P}(V_{AD} \mid E_{BC}^c) = q + p(1 - (1 - q^2)^2)$$

Part d)

We apply the Law of Total Probability by partitioning the space on the direct edge E_{AD} and its complement E_{AD}^c :

$$\mathbb{P}(V_{AD} \mid E_{AB}^c) = \mathbb{P}(V_{AD} \mid E_{AD} \cap E_{AB}^c) \mathbb{P}(E_{AD} \mid E_{AB}^c) + \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{AB}^c) \mathbb{P}(E_{AD}^c \mid E_{AB}^c) \quad (5)$$

By the independence of the individual edges, we have $\mathbb{P}(E_{AD} \mid E_{AB}^c) = \mathbb{P}(E_{AD}) = q$ and $\mathbb{P}(E_{AD}^c \mid E_{AB}^c) = \mathbb{P}(E_{AD}^c) = p$. Furthermore, as established previously, $\mathbb{P}(V_{AD} \mid E_{AD} \cap E_{AB}^c) = 1$.

If $E_{AD}^c \cap E_{AB}^c$ occurs, the rumor cannot travel directly to D or through B initially. Therefore, it must travel through C (requiring E_{AC}). From C , it can reach D directly (E_{CD}) or via B ($E_{CB} \cap E_{BD}$):

$$\mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{AB}^c) = \mathbb{P}(E_{AC} \cap (E_{CD} \cup (E_{CB} \cap E_{BD})) \mid E_{AD}^c \cap E_{AB}^c)$$

Because the edges required for this path are disjoint from the conditioned edges, they are independent. This allows us to drop the conditioning:

$$\mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{AB}^c) = \mathbb{P}(E_{AC}) \cdot \mathbb{P}(E_{CD} \cup (E_{CB} \cap E_{BD}))$$

By the independence of the disjoint edge sets and the Inclusion-Exclusion Principle:

$$\begin{aligned} \mathbb{P}(V_{AD} \mid E_{AD}^c \cap E_{AB}^c) &= \mathbb{P}(E_{AC}) [\mathbb{P}(E_{CD}) + \mathbb{P}(E_{CB} \cap E_{BD}) - \mathbb{P}(E_{CD} \cap E_{CB} \cap E_{BD})] \\ &= q(q + q^2 - q^3) \end{aligned}$$

Substituting these values back into equation 5, we obtain:

$$\begin{aligned} \mathbb{P}(V_{AD} \mid E_{AB}^c) &= 1 \cdot q + p [q(q + q^2 - q^3)] \\ &= q + pq(q + q^2 - q^3) \end{aligned}$$